# SOFTWARE REQUIREMENTS SPECIFICATION

Project: Smart Contract Enabled Pharmaceutical Supply Chain System
Vendor: PharmaTrust Software Development Team
Customer: Medicare Plus
Template Used: IEEE 830-1998

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## Declaration of AI Use & Plagiarism

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# 1. INTRODUCTION

## 1.1. Purpose

The purpose of the “Smart Contract Enabled Pharmaceutical Supply Chain System for Anti-Counterfeiting and Product Traceability” is to establish a secure, transparent, and tamper-resistant digital infrastructure that enables end-to-end traceability of pharmaceutical products across the entire supply chain lifecycle.

Medicare Plus currently operates within a fragmented and partially paper-based supply chain environment, where suppliers, manufacturers, wholesalers, distributors, and transporters maintain separate records. This fragmentation increases operational inefficiencies, creates reconciliation delays, and exposes the organisation to risks including counterfeit product infiltration, unauthorised diversion, inventory discrepancies, and regulatory non-compliance.

This project aims to address these challenges by implementing a blockchain-based system that:

- Records all critical supply chain transactions immutably using smart contracts
- Automates order lifecycle processes to reduce manual intervention and administrative overhead
- Enables real-time verification of medicine authenticity and origin
- Detects and prevents tampering through cryptographic hash validation
- Provides a single trusted source of truth accessible to authorised stakeholders
- Ensures audit readiness and regulatory compliance under applicable pharmaceutical governance frameworks

By integrating smart contract automation with secure role-based access control and traceable transaction logging, the system will enhance operational efficiency, strengthen supply chain integrity, and protect patient safety through reliable product provenance verification.

The intended outcome of this project is a regulator-ready, anti-counterfeiting, and audit-compliant pharmaceutical traceability platform that supports sustainable operational growth and reinforces organisational trust across all participating stakeholders.

## 1.2. Document Conventions

| Priority Level | Classification |
|---|---|
| **P0** | High |
| **P1** | Medium |
| **P2** | Low |

| Category | Description |
|---|---|
| **SHALL** | Mandatory requirement |
| **SHOULD** | Strong recommendation |
| **MAY** | Optional behavior |

| Category | Prefix |
|---|---|
| Functional Requirement | FR-x.x |
| Non-Functional Requirement | NFR-x |
| Quality Factor | QF-x |
| Assumption | A-x |
| Constraint | C-x |
| Change Mitigation | CM-x |
| Technical Requirement | TR-x |

### 1.3. Intended Audience

**Primary Audience (Direct Users)**

Customer Stakeholders (MediCare Plus Network)

- IT Department: System administrators, database administrators, and IT security personnel responsible for system deployment, configuration, and ongoing maintenance
- Supply Chain Management: Operations managers overseeing pharmaceutical procurement, distribution, and logistics coordination
- Compliance & Legal: Regulatory compliance officers and legal advisors ensuring adherence to HSA, MOH, PDPC, and ISO standards
- Executive Management: Chief Information Officer (CIO), Chief Operating Officer (COO), and senior leadership evaluating strategic alignment and ROI
- Quality Assurance: QA personnel responsible for product integrity verification and audit trail validation
- Finance Department: Financial controllers and procurement officers managing vendor contracts, payment schedules, and cost tracking

Vendor Stakeholders (PharmaTrust Distributors Ltd.)

- Development Team: Software engineers, blockchain developers, and smart contract programmers implementing system functionality
- Technical Architects: Solution architects and system designers responsible for technical design decisions and integration planning
- Project Management: Project managers, business analysts, and requirements engineers coordinating delivery milestones and stakeholder communication
- DevOps & Infrastructure: Cloud engineers and DevOps specialists managing Azure deployment, CI/CD pipelines, and production environment configuration

**Secondary Audience (Review & Approval)**

Quality Assurance & Testing

- QA Engineers: Test engineers conducting functional, integration, performance, and security testing
- UAT Coordinators: User acceptance testing facilitators managing stakeholder validation sessions
- Security Auditors: Cybersecurity specialists performing penetration testing and vulnerability assessments

Regulatory & Compliance Reviewers

- HSA Reviewers: Health Sciences Authority inspectors evaluating GDP and GMP compliance
- MOH Reviewers: Ministry of Health officials assessing healthcare services compliance and data security
- PDPC Auditors: Personal Data Protection Commission auditors reviewing data protection measures and consent management
- ISO Certification Bodies: ISO 9001 and ISO 27001 auditors assessing quality management and information security controls

External Stakeholders

- Supply Chain Partners: Suppliers, manufacturers, wholesalers, distributors, and third-party logistics providers who will interact with the system
- Healthcare Providers: Hospital pharmacies, clinics, and healthcare institutions integrating with the platform
- Legal Advisors: External legal counsel reviewing contractual obligations, liability clauses, and intellectual property considerations

### 1.4. Project Scope

This system will provide blockchain-based supply chain traceability that tracks every pharmaceutical product from raw material procurement through to final delivery across Singapore's healthcare network, preventing counterfeit medicine entry by creating cryptographically-secured, immutable custody records that make product forgery computationally infeasible.

Automate the entire order lifecycle through smart contracts deployed on the Stratis blockchain to eliminate manual processing overhead and enforce business rules aligned with Singapore regulatory requirements, providing audit-ready transaction history with Singapore Standard Time (SST) timestamps and digital signatures to support compliance reporting for HSA regulatory inspections, MOH Healthcare Services Act audits, and PDPC data protection assessments. While enforcing secure role-based access control (RBAC) with multi-factor authentication (MFA) compliant with the Personal Data Protection Act 2012 and MOH Cyber and Data Security Guidelines 2023 to ensure that administrators, suppliers, manufacturers, wholesalers, distributors, transporters, and customers can only access functionality which is appropriate to their role while maintaining complete audit trails of all system interactions.

#### 1.4.1. In-Scope Deliverables

**Core System Functionality:**

Blockchain Infrastructure

- Implementation of Stratis blockchain platform with Cirrus Core sidechain integration
- Deployment of Proof-of-Authority (PoA) consensus mechanism with minimum 5 validator nodes
- SHA-256/SHA-512 cryptographic hash functions for block integrity verification
- Smart contract deployment and management framework using C# on .NET 6.0

User Role Management

- Administrative Module: User account approval workflows, smart contract assignment, tamper detection alerts, and system monitoring dashboards
- Supplier Module: Raw material order management, inventory tracking, transporter assignment, and shipment coordination
- Manufacturer Module: Medicine production recording, raw material procurement, order verification, and NDC code serialization
- Wholesaler Module: Order processing from manufacturers, distributor fulfillment, delivery coordination, and high-value shipment dual authorization
- Distributor Module: Wholesaler order management, customer order processing, last-mile delivery coordination, and product authenticity verification
- Customer Module: Medicine search and ordering, QR code-based source verification, order tracking, and supply chain history viewing
- Transporter Module: Shipment assignment acceptance, GPS location updates, custody transfer confirmation, and delivery status reporting

Authentication & Security

- Multi-factor authentication (MFA) for all user accounts with SMS/email OTP support
- Role-based access control (RBAC) with the principle of least privilege enforcement
- Public/private key cryptography for digital signatures on blockchain transactions
- AES-256 encryption for data at rest and TLS 1.3 for data in transit

Data Management & Storage

- Microsoft SQL Server for off-chain operational data (user profiles, product catalogs, order metadata)
- Blockchain ledger for transaction hashes, custody records, and immutable audit trails
- 7-year data retention policy compliant with the PDPC Personal Data Protection Act 2012
- Automated daily backups with point-in-time recovery capability

User Interface & Experience

- Responsive web interface compatible with Chrome 100+, Firefox 100+, Edge 100+, Safari 15+
- Mobile-responsive design supporting iOS 15+ and Android 11+ devices (minimum 1024x768 resolution)
- Desktop application support for Windows 10/11, macOS 12+, and modern Linux distributions
- Accessibility compliance with WCAG 2.1 Level AA standards

**Integration Capabilities**

External System Integrations

- Epic EHR Integration: HL7/FHIR R4 API for prescription validation and medication administration records
- SAP ERP Integration: REST API with OAuth 2.0 for inventory management, financial transactions, and procurement workflows
- Omnicell Automated Dispensing: Vendor API connector for medication dispensing verification
- FDA Sentinel System: HTTPS data submission for adverse event reporting

Regulatory Compliance Features

- HSA Good Distribution Practice (GDP): Full traceability with batch numbers, expiry dates, quantities, supplier/purchaser documentation
- HSA Health Products Act 2007: Registration verification for all therapeutic products and license validation for manufacturers/importers/wholesalers
- GMP Compliance: Manufacturer certification verification and production quality control tracking
- MOH Healthcare Services Act 2020: Confidentiality of patient/medical information and healthcare institution licensing compliance
- MOH Cyber & Data Security Guidelines 2023: Cybersecurity controls for healthcare IT systems
- PDPC Personal Data Protection Act 2012: Consent management, purpose limitation, data protection, breach notification, and 7-year retention
- IMDA Electronic Transactions Act: Legal validity of smart contracts and electronic signatures
- ISO 9001 (Quality Management): Quality assurance processes and continuous improvement tracking
- ISO 27001 (Information Security): Information security management system (ISMS) controls and risk assessment

**Operational Capabilities**

Anti-Counterfeiting Measures

- QR code generation with cryptographic signatures for each product unit
- Blockchain-based source verification allowing customers to trace complete custody chain
- Tamper detection through hash validation at each custody transfer point
- Real-time alerts for unauthorised product entries or manipulation attempts

Process Automation

- Smart contract-driven order lifecycle automation (creation, verification, approval, fulfillment)
- Automated custody transfer recording with digital signatures from both parties
- Workflow enforcement for high-value shipment dual authorization (>$50,000 SGD threshold)
- Automated notification system via webhooks for status changes and exception handling

Audit & Reporting

- Immutable audit trail for all supply chain transactions with UTC timestamps
- Compliance report generation for HSA, MOH, and PDPC regulatory submissions
- Transaction history export in CSV, PDF, and Excel formats
- Real-time dashboards for supply chain visibility and performance metrics

**Performance & Scalability**

Performance Requirements

- Initial page load time: <3 seconds for desktop, <5 seconds for mobile on 4G connection
- Subsequent page interactions: <1 second response time for standard operations
- Blockchain transaction confirmation: <30 seconds for standard transactions, <60 seconds for complex smart contract executions
- Database query performance: 95% of queries complete in <500ms; complex analytical queries <5 seconds

Scalability Requirements

- Support for a minimum of 500 concurrent users without performance degradation
- Handle 2,500+ pharmaceutical transactions per day with a peak capacity of 500 transactions per hour
- Horizontal scaling capability for future volume growth (up to 10,000 transactions/day)

**Deployment & Infrastructure**

Hosting Environment

- Microsoft Azure cloud infrastructure with 99.9% uptime SLA
- Production environment in Southeast Asia region (Singapore) region with backup replication to East Asia region (Hong Kong) for disaster recovery.
- Separate development, staging, and UAT environments with production-like configurations to support testing, validation, and deployment readiness.

Maintenance & Support

- Scheduled maintenance windows: Maximum 4 hours/month during off-peak hours (2:00-6:00 AM EST)
- CI/CD pipeline using Azure DevOps or GitHub Actions for automated testing and deployment
- Blue-green deployment strategy for zero-downtime upgrades with 15-minute rollback capability
- Security patches applied within 48 hours of release; feature updates deployed monthly

**1.4.2. Out-of-Scope Items**

The following capabilities are explicitly excluded from the current phase:

Phase 2 (Future Enhancements)

- Temperature monitoring and cold chain management Direct payment gateway integration (Stripe, PayPal) for B2C transactions
- Mobile native applications (iOS/Android apps), phase 1 delivers mobile-responsive web only
- AI/ML-powered demand forecasting or inventory optimization
- Real-time video streaming for shipment monitoring
- Integration with additional EHR systems beyond Epic (e.g., Cerner, Allscripts)
- Blockchain interoperability with other pharmaceutical traceability platforms (e.g., MediLedger, IBM Food Trust)

Explicitly Non-Functional

- Physical product authentication (e.g., RFID tags, NFC chips, holographic labels) system provides digital verification only
- Physical security measures (warehouse access control, tamper-evident packaging) responsibility of supply chain partners
- Customs clearance and international shipping documentation are out of the system scope
- Clinical trial management or pharmacovigilance beyond adverse event reporting
- Patient medication adherence tracking or reminders
- Drug interaction checking or clinical decision support

Third-Party Responsibilities

- Hardware procurement for blockchain validator nodes (customer responsibility)
- GPS tracking hardware installation in transporter vehicles
- IoT sensor deployment for temperature monitoring (deferred to Phase 2)
- Training delivery logistics (training materials provided; delivery coordinated by customer)
- User onboarding and change management execution (vendor provides support; customer leads execution)

Excluded Regulatory Jurisdictions

- FDA compliance for US market (system is Singapore HSA/MOH focused)
- EMA compliance for European market
- PMDA compliance for Japanese market
- Any regulatory frameworks outside Singapore jurisdiction unless explicitly contracted

**1.4.3. System Boundaries**

Geographic Scope

- Primary deployment: Singapore-based operations for MediCare Plus Network
- Data residency: All data stored in compliance with Singapore PDPC requirements
- Regulatory framework: Singapore HSA, MOH, PDPC, IMDA, ISO standards

Organizational Scope

- Included: MediCare Plus and its authorized supply chain partners (suppliers, manufacturers, wholesalers, distributors, transporters)
- Included: Healthcare providers with Epic EHR integration
- Excluded: Retail pharmacy point-of-sale systems (unless specifically contracted)
- Excluded: Patient-facing mobile apps or patient portals (Phase 1 focuses on B2B supply chain)

Technical Scope

- Included: Web-based user interface, blockchain ledger, off-chain databases, API integrations
- Excluded: Legacy system migration (data migration tools provided; migration execution is customer responsibility)
- Excluded: On-premises deployment options (cloud-only in Phase 1)

**1.4.4. Success Criteria**

The project will be considered successful when:

- Functional Completeness: All FR-1.1 through FR-4.4 implemented and verified through UAT
- Performance Targets Met: 95% of blockchain transactions recorded successfully (SMART goal), manual processing reduced by 60%, page load times meet TR-40 specifications
- Security & Compliance: Pass external security audit, demonstrate compliance with HSA GDP, MOH CSDS, PDPC PDPA, ISO 27001
- User Acceptance: Achieve 80% positive feedback during UAT (SMART goal), 100% of users able to retrieve medicine traceability information
- Regulatory Approval: Receive sign-off from MediCare Plus compliance officers and HSA reviewers (if applicable)
- Production Deployment: System deployed to production with 99.9% uptime SLA, zero critical defects in first 30 days post-launch

### 1.5 References

[1] Personal Data Protection Commission (PDPC), *Personal Data Protection Act 2012*, Singapore Statutes Online, Singapore, 2012.

[2] IEEE, *IEEE Recommended Practice for Software Requirements Specifications*, IEEE Std 830-1998, IEEE, 1998.

[3] D. Le Vie Jr., "Writing Software Requirements Specifications," *TechWhirl*, [Online]. Available: https://techwhirl.com/writing-software-requirements-specifications/.

[4] Le Vie, D., Jr. (n.d.). *Writing software requirements specifications*. TechWhirl.

## 2. OVERALL DESCRIPTION

### 2.1. Product Perspective

System architecture:

- ASP.NET (C#) backend
- MSSQL database
- Stratis Smart Contract blockchain
- Cloud Windows Server

Architecture Highlights (**Fig 3**):

- Five-Layer Architecture: Clear separation between Presentation, Application, Data, Blockchain, and Infrastructure layers
- Hybrid Storage: Off-chain MSSQL for operational data + On-chain blockchain for immutable transactions
- Scalability: 2-10 load-balanced servers, auto-scaling, 500+ concurrent users supported
- Performance: <3s page load, <30s blockchain confirmation, 95% queries <500ms
- Security: AES-256, TLS 1.3, MFA, RBAC, public/private key cryptography
- Availability: 99.9% up time
- Compliance: HSA GDP/GMP, MOH CSDS, PDPC PDPA, ISO 9001/27001 aligned

### 2.2. Product Features

High-level system features:

- User & Role Management
- Smart Contract Automation
- Order Lifecycle Management
- Blockchain Logging
- Product Verification
- Audit Reporting

### 2.3. User Classes and Characteristics

- Admin – Full system control
- Supplier – Raw material management
- Manufacturer – Production & ordering
- Wholesaler – Medicine procurement
- Distributor – Distribution management
- Transporter – Logistics execution
- Customer – Verification & purchase

### 2.4. Operating Environment (Latest Technology)

- Cloud-based deployment
- Windows Server 2022
- MSSQL 2019FR-
- Stratis Smart Contract platform

### 2.5. Design & Implementation Constraints

C-1 Backend must use ASP.NET (C#)
C-2 Blockchain platform fixed to Stratis
C-3 Database must use MSSQL
C-4 Must comply with PDPC & HSA

### 2.6. User Documentation

- Online help guide
- Role-specific manuals
- Administrator guide

### 2.7. Assumptions and Dependencies

A-1 Internet connectivity available
A-2 Blockchain network uptime ≥ 99%
A-3 Regulatory framework stable

## 3. SYSTEM FEATURES

**Feature: Blockchain Infrastructure**
**Description & Priority**
High priority. Core traceability mechanism.

**Stimulus/Response**
Stimulus: Order approval
Response: Smart contract executes and logs blockchain entry.

### 3.1. Functional Requirements

FR-1.1 The system shall **record** all approved supply-chain transactions on the blockchain ledger.
FR-1.2 The system shall **validate** blockchain hash integrity for each recorded transaction.
FR-1.3 The system shall **execute** smart contracts to automate order state transitions based on authorized events.
FR-1.4 The system shall allow authorized users to **view** the blockchain transaction record associated with an order

### 3.2. Feature: User & Role Management

FR-2.1 The system shall allow an Admin to **create** user accounts and assign roles and smart contract accounts.
FR-2.2 The system shall allow an Admin to **view** user profiles, roles, and account status
FR-2.3 The system shall allow an Admin to **update** user roles, permissions, and account status.
FR-2.4 The system shall enforce role-based access control (RBAC) for all functions and data access.
FR-2.5 The system shall enforce password minimum length, complexity rules, account lockout thresholds, and secure cryptographic hashing.
FR-2.6 The system shall enforce multi-factor authentication (MFA) for all administrative accounts.

### 3.3. Feature: Order Lifecycle Management

FR-3.1 The system shall allow Manufacturers to **create** raw material orders.
FR-3.2 The system shall allow Wholesalers and Distributors to **create** medicine orders.

FR-3.3 The system shall allow authorized users to **view** order details, order history, and current status.
FR-3.4 The system shall allow authorized roles to **update** order status based on workflow rules.
FR-3.5 The system shall allow each order to be assigned to **one transporter** at a time.
FR-3.6 The system shall validate GMP certification before accepting a medicine production entry from a Manufacturer.
FR-3.7 The system shall record GDP-compliant audit logs for Wholesaler order processing actions.
FR-3.8 The system shall allow Customers to place medicine orders only after successful prescription validation.
FR-3.9 The system shall allow Transporters to submit periodic GPS updates during shipment transit.
FR-3.10 The system shall support batch number and lot number tracking for recall management.

### 3.4. Feature: Product Verification

FR-4.1 The system shall allow Customers to **retrieve and view** the full product lifecycle trace (end-to-end traceability).
FR-4.2 The system shall display the blockchain transaction ID and block number for each traceability record.
FR-4.3 The system shall validate National Drug Code entries against the HSA registry when product master data is created or updated.
FR-4.4 The system shall generate regulatory reports aligned with MOH, HSA, and PDPC requirements and retain generated reports for seven years.
FR-4.5 The system shall integrate with Epic EHR using HL7/FHIR R4 APIs solely for prescription validation.
FR-4.6 The system shall synchronise inventory and financial data with SAP ERP OAuth 2.0 REST API daily

The process can be better visualised with a **simplified sequence diagram** (Fig 2.) to give a simple idea of user operations. To show how the system can be used.

### 3.5. Wireframe Reference

The customer's Project Specification includes preliminary wireframes representing the expected layout, navigation, and role-based interaction flow of the proposed system. These wireframes (Fig 5.1-5.6) are referenced in this SRS as supporting interface artefacts to preserve alignment with the customer's original intent. They shall be treated as baseline visual guidance, and may be refined during subsequent design and usability validation activities.

## 4. EXTERNAL INTERFACE REQUIREMENTS

### 4.1. User Interfaces

- Wireframe Login Screen
- Wireframe Customer Dashboard
- Wireframe Medicine Search & Source Verification
- Wireframe Order Tracking
- Wireframe Admin Dashboard
- Wireframe Supplier/Manufacturer Portal

### 4.2. Hardware Interfaces

- Standard web browser client devices.
- Blockchain validator nodes (minimum 5 PoA nodes)
- GPS tracking hardware for transporters
- Cloud virtual machines (Azure infrastructure)

### 4.3. Software Interfaces

- MSSQL
- Stratis Smart Contract
- Authentication module

### 4.4. Communications Interfaces

- HTTPS secure communication
- Blockchain network communication protocol

## 5. OTHER NONFUNCTIONAL REQUIREMENTS

### 5.1. Performance Requirements (Efficiency/performance/throughput/latency)

NFR-1 The system shall load the initial page within 3 seconds on desktop; 5 seconds on mobile (TR-40)
NFR-2 The system shall confirm Blockchain transaction within 10 seconds for standard operations (TR-44)
NFR-3 The system shall support a minimum of 500 concurrent users without performance degradation
NFR-4 The system shall shall complete 95% of API responses within 2 seconds (TR-16)
NFR-5 The system shall handle a peak transaction volume of 500 transactions per hour (TR-43)
NFR-6 The system shall respond subsequent page interactions within 1 second (TR-41)
NFR-7 The system shall shall complete 95% of database queries within 500ms
NFR-8 The system shall update orders within 3 seconds

### 5.2. Safety Requirements (Integrity, encryption, access control, hashing, tamper prevention)

NFR-9 The system shall reject any create/update request to traceability data unless the request is authenticated and authorized by RBAC, and the attempt shall be recorded in audit logs.

### 5.3. Security Requirements

NFR-10 The system shall encrypt data at rest using AES-256 (TR-23)
NFR-11 The system shall protect data in transit by TLS 1.3 (TR-24)
NFR-12 The system shall shall enforce Role-based access control (RBAC) on every system request (TR-25)
NFR-13 The system shall retain audit logs for a minimum of seven years (TR-26)
NFR-14 The system shall store passwords using salted, adaptive one-way hashing and shall not store passwords in reversible form.
NFR-15 The system shall use SHA-256 / SHA-512 for all blockchain transactions to ensure block integrity verification
NFR-16 The system shall maintain blockchain hash integrity for each recorded transaction has to be validated within 5 seconds

### 5.4. Availability & Continuity (Reliability, uptime/backup/RTO/RPO)

NFR-17 The system shall have an update of ≥ 99.9% monthly (TR-01)
NFR-18 The system shall implement geographic redundancy across at least two Azure regions (TR-02)
NFR-19 The system shall have Backup RPO ≤ 24 hours; RTO shall be ≤ 4 hours (TR-27)

### 5.5. Software Quality Attributes

This system adopts a customised application of McCall's Software Quality Framework, tailored to the operational, regulatory, and integrity-critical context of a blockchain-enabled pharmaceutical supply chain system. The selected quality factors are derived from McCall's Operation and Revision perspectives and are mapped directly to system risks, regulatory expectations, and smart contract deployment considerations. Quality factor names have been adapted from McCall's standard framework to reflect the regulatory, anti-counterfeiting and audit-critical context of this system.

#### 5.5.1. QF-1 Supply Chain Logic Fidelity

**Definition**

Correctness is the extent to which the system accurately implements all specified functional requirements and enforces business rules governing pharmaceutical supply chain traceability.

**Project Context Relevance**
In this system, correctness is critical because incorrect logic could allow counterfeit product insertion:
- Each blockchain transaction represents a legally auditable record.
- Order state transitions must strictly follow supply chain logic.
- Incorrect traceability may result in regulatory violation or patient safety risk.
- Smart contracts must enforce agreed business rules without deviation.

**Measurable Criteria**
- ≥ 95% functional test case coverage of all FR clauses.
- 100% validation of order state transition rules.
- Zero unresolved discrepancies between order database state and blockchain state.
- 100% successful hash validation during traceability lookup tests.

**Linked Requirements**
- FR-1.1 Transaction Recording
- FR-3.4 Order Status Update
- FR-4.1 Source Verification

### 5.5.2. QF-2 Traceability Continuity
**Definition**
Reliability is the extent to which the system performs its intended functions consistently without failure under defined operational conditions.

**Project Context Relevance**
The pharmaceutical supply chain operates continuously. System downtime may:
- Delay medicine distribution
- Prevent order confirmation
- Block product verification
- Interrupt regulatory audit processes

Blockchain interaction failures or smart contract execution errors could compromise traceability continuity.

**Measurable Criteria**
- System uptime shall be ≥ 99.9% monthly.
- Blockchain write confirmation success rate ≥ 99.5%.
- Mean Time To Recovery (MTTR) ≤ 2 hours.
- No critical system failure exceeding 1 hour without mitigation.

**Linked Requirements**
- NFR-15 Uptime Requirement
- FR-1.3 Smart Contract Execution
- CM-4 Contingency Strategy

### 5.5.3. QF-3 Tamper Resistance
**Definition**
Integrity is the extent to which the system prevents unauthorised access, data tampering, and fraudulent modification of pharmaceutical transaction records.

**Project Context Relevance**
Integrity is the core objective of this system because:
- Counterfeit medicines represent both financial and patient safety risks.
- Blockchain immutability must be preserved.

- Smart contract logic must not be bypassed.
- Only authorised stakeholders may create or validate supply chain records.

Integrity ensures the system functions as a “single trusted source of truth.”

**Measurable Criteria**

- 100% enforcement of role-based access control.
- Zero successful unauthorised access attempts during penetration testing.
- 100% hash consistency validation for blockchain trace requests.
- Mandatory authentication for all state-changing operations.

**Linked Requirements**

- FR-1.2 Tamper Validation
- FR-2.4 Role-Based Access Control
- NFR-14 SHA-256 Hashing

#### 5.5.4. QF-4 Regulatory Adaptability

**Definition**

Maintainability is the extent to which the system can be modified, extended, or corrected without introducing unintended defects or disrupting existing traceability records.

**Project Context Relevance**

In a regulatory-sensitive environment:

- Business rules may change due to new compliance regulations.
- Smart contract logic may require updates.
- Additional stakeholder roles may be introduced.
- Audit requirements may evolve.

Given blockchain immutability, changes must be carefully managed to avoid historical data corruption.

**Measurable Criteria**

- Smart contracts version-controlled with documented upgrade path.
- Modular architecture separating:
    - Business logic
    - Blockchain interaction layer
    - User interface layer
- All change requests processed through CCB approvals
- Backward compatibility is maintained unless formally deprecated.

**Linked Requirements**

- CM-3 Smart Contract Versioning
- CM-2 Change Classification
- iRTM Update Protocol

#### 5.5.5. QF-5 Audit Verifiability

**Definition**

Testability is the extent to which the system allows effective verification, diagnostics, and audit simulation to confirm compliance and correctness.

**Project Context Relevance**

Testability is critical because:

- Regulatory audits require reproducible traceability evidence.
- Smart contracts must be validated before deployment.

- Hash consistency must be verifiable.
- Regression testing must validate that blockchain logic remains stable after changes.

Without strong testability, regulatory approval may fail.

**Measurable Criteria**

- Automated test coverage ≥ 90%.
- Dedicated staging blockchain network for smart contract testing.
- Audit simulation execution prior to production release.
- Log retrieval time ≤ 1 minute.
- All FR and NFR clauses were mapped to test cases via iRTM.

**Linked Requirements**

- FR-4.2 Audit Trail Display
- NFR-12 Compliance Logs
- iRTM Matrix

**Quality Mapping Summary**

| Quality Factor | Project Risk Addressed |
|---|---|
| Correctness | Incorrect supply chain logic |
| Reliability | Supply disruption |
| Integrity | Counterfeit & tampering |
| Maintainability | Regulatory change risk |
| Testability | Audit & compliance failure |

## 6. OTHER REQUIREMENTS

### 6.1. Inceptive Requirements Traceability Matrix (iRTM)

The fleshed-out **iRTM** (Fig 4.) can be found under the **FIGURES** section, the following tables are for association with the proper sequential audit of the respective requirements to show a sequence from Project Specification to Analysis Report to the current Software Specification Report. Along with a **context diagram** (Fig 1.), which shows flow of data, with the associated Functional Requirements (FR)

**Blockchain Infrastructure Requirements**

| PS ID | AR ID | SRS ID | Type | McCall Factor | Design Ref | Test Ref | Change Ref |
|---|---|---|---|---|---|---|---|
| PS-BR-01 Blockchain recording | AR-FR-1.1 | FR-1.1 | FR | Correctness | UC-01, SD-01 | TC-01 | CM-2 |
| PS-BR-02 Tamper detection | AR-FR-1.2 | FR-1.2 | FR | Integrity | SD-02 | TC-02 | CM-3 |
| PS-BR-03 Smart contract automation | AR-FR-1.3 | FR-1.3 | FR | Reliability | SD-03 | TC-03 | CM-3 |

**User & Role Management Requirements**

| PS ID | AR ID | SRS ID | Type | McCall Factor | Design Ref | Test Ref | Change Ref |
|---|---|---|---|---|---|---|---|
| PS-BR-04 Account creation | AR-FR-2.1 | FR-2.1 | FR | Correctness | UC-02 | TC-04 | CM-1 |
| PS-BR-05 Role access control | AR-FR-2.4 | FR-2.4 | FR | Integrity | UC-03 | TC-05 | CM-1 |
| PS-BR-06 Customer approval | AR-FR-2.3 | FR-2.3 | FR | Integrity | SD-04 | TC-06 | CM-2 |

**Order Lifecycle Requirements**

| PS ID | AR ID | SRS ID | Type | McCall Factor | Design Ref | Test Ref | Change Ref |
|---|---|---|---|---|---|---|---|
| PS-BR-07 Raw material orders | AR-FR-3.1 | FR-3.1 | FR | Correctness | UC-04 | TC-07 | CM-2 |
| PS-BR-08 Medicine orders | AR-FR-3.2 | FR-3.2 | FR | Correctness | UC-05 | TC-08 | CM-2 |
| PS-BR-09 Status update | AR-FR-3.4 | FR-3.4 | FR | Reliability | SD-05 | TC-09 | CM-3 |
| PS-BR-10 Transport assignment | AR-FR-3.5 | FR-3.5 | FR | Correctness | UC-06 | TC-10 | CM-2 |

**Product Verification Requirements**

| PS ID | AR ID | SRS ID | Type | McCall Factor | Design Ref | Test Ref | Change Ref |
|---|---|---|---|---|---|---|---|
| PS-BR-11 Source lookup | AR-FR-4.1 | FR-4.1 | FR | Integrity | SD-06 | TC-11 | CM-2 |
| PS-BR-12 Audit display | AR-FR-4.2 | FR-4.2 | FR | Testability | SD-07 | TC-12 | CM-3 |

**Non-Functional Requirements (Security & Compliance)**

| PS ID | AR ID | SRS ID | Type | McCall Factor | Design Ref | Test Ref | Change Ref |
|---|---|---|---|---|---|---|---|
| PS-TR-01 Encryption | AR-(NFR-9& NFR-10) | NFR-9& NFR-10 | NFR | Integrity | Arch-01 | TC-13 | CM-1 |
| PS-TR-02 Hash algorithm | AR-NFR-14 | NFR-14 | NFR | Integrity | Arch-02 | TC-14 | CM-3 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| PS-TR-03 Performance | AR-(NFR-1 to NFR-7) | NFR-1 to NFR-7 | NFR | Efficiency | Perf-01 | TC-15 | CM-2 |
| PS-TR-04 Uptime | AR-NFR-15 | NFR-15 | NFR | Reliability | Arch-03 | TC-16 | CM-4 |
| PS-TR-05 Compliance logs | AR-NFR-12 | NFR-12 | NFR | Testability | SD-08 | TC-17 | CM-4 |

### 6.2. Change Mitigation Strategy

#### 6.2.1. CM-1 Change Control Board (CCB)

All proposed changes shall be reviewed and approved by the *Change Control Board (CCB)*, formed with representation from both customer and vendor teams to ensure governance over high-criticality changes, particularly those involving smart contracts and compliance. The customer side shall be composed of three representatives: the IT Lead, Compliance/Regulatory Officer, and Operations/Supply Chain Representative. The vendor side shall be composed of 5: the Project Director or Project Manager as Chair, the Solution/Blockchain Architect whose attendance is mandatory for smart contract changes, the Lead Backend Engineer, the QA Lead, and the DevOps/Infrastructure Lead.

The CCB shall review all change requests and confirm their business and regulatory rationale, assessing impact on functional behaviour, quality factors, regulatory compliance, data integrity, blockchain immutability, and system uptime. It shall classify the severity of each change and decide to approve, reject, or defer accordingly, while defining the required testing scope, acceptance of evidence, and ensuring all SRS and iRTM updates are completed prior to release. No change affecting smart contracts or traceability logic shall bypass the CCB under any circumstance.

#### 6.2.2. CM-2 Change Classification Framework

Changes are categorised to determine control intensity.

| Change Type | Examples | Control Level |
|---|---|---|
| Minor Change | • UI text updates<br>• Non-logic configuration changes<br>• Report formatting adjustments | • Vendor internal review<br>• QA validation<br>• No blockchain redeployment required |
| Major Functional Change | • New stakeholder role<br>• New order state transition<br>• Additional verification step<br>• Modified traceability logic | • Full CCB approval<br>• Impact analysis on FR clauses, McCall quality attributes, and iRTM mappings<br>• Regression testing required<br>• Deployment to staging before production |
| Regulatory / Compliance Change *(High Criticality)* | • New HSA reporting requirement<br>• PDPC data retention amendment<br>• Audit evidence restructuring<br>• Mandated traceability extension | • Priority review within 48 hours<br>• Compliance officer sign-off required<br>• Formal documentation update<br>• Version-controlled release<br>• Mandatory regression and audit simulation |

#### 6.2.3. CM-3 Smart Contract Versioning Strategy

Smart contracts are immutable once deployed. Therefore, special mitigation controls apply. This prevents irreversible blockchain corruption.

| Strategies | Details |
|---|---|
| **Versioning Model** | • Each smart contract shall carry a version identifier<br>• Previous versions remain archived<br>• Version mapping maintained in iRTM |
| **Upgrade Procedure** | • New contract deployed to staging network<br>• Integration and regression tests executed<br>• Compatibility check with existing blockchain records<br>• Controlled migration of state (if required)<br>• Production deployment after CCB approval |
| **Rollback Strategy** | • The system shall suspend the affected transaction type<br>• Previous contract version reactivated within 24 hours<br>• Incident logged in audit trail<br>• Root cause analysis conducted |

#### 6.2.4. CM-4 Blockchain Risk Contingency

| Contingencies | Details |
|---|---|
| **Fork Handling Procedure** | • Transactions temporarily paused<br>• Fork integrity assessed<br>• Longest valid chain adopted<br>• Audit validation executed before resuming operations |
| **Hash Mismatch Detection** | • Transaction flagged<br>• Admin notified<br>• Forensic review initiated<br>• Affected user access is temporarily restricted |

#### 6.2.5. CM-5 Data Recovery & Continuity

**Database Recovery**

- Full backup every 6 hours.
- Incremental backup every 30 minutes.
- Recovery time objective (RTO) ≤ 2 hours.
- Recovery point objective (RPO) ≤ 30 minutes.

**Incident Response**

- Critical system incident response ≤ 1 hour.
- Dedicated incident escalation chain.
- Incident log permanently stored.

#### 6.2.6. CM-6 Quality & Traceability Safeguards

All approved changes must:

1. Update iRTM mappings.
2. Re-evaluate McCall quality factors:
    - Correctness
    - Reliability
    - Integrity
    - Maintainability
    - Testability
3. Undergo regression test coverage ≥ 90%.
4. Update SRS version number.
5. Maintain backward compatibility unless formally deprecated.

#### 6.2.7. CM-7 Supply Chain Continuity Protection

Because the system supports pharmaceutical logistics:

- No deployment shall occur during the peak supply window.
- Rollout windows must be scheduled.
- Emergency patch protocol defined.
- Transaction buffering enabled during short downtime.

This ensures no medicine distribution is interrupted.

#### 6.2.8. CM-8 Audit Trail Preservation

All changes must:

- Be logged with a timestamp.
- Include change rationale.
- Include approving authority.
- Be retrievable for regulatory audit.
- Audit logs shall be immutable.

#### 6.2.9. Ownership (Vendor accountable roles)

| Change Mitigation | Stakeholders |
|---|---|
| **CM-1 CCB governance** | - Project Director / PM (Accountable)<br>- Requirements Engineer (Secretary + change log) |
| **CM-2 Change classification** | - Requirements Engineer / BA (Accountable)<br>- QA Lead (consulted for test impact) |
| **CM-3 Smart contract versioning** | - Blockchain Architect (Accountable)<br>- Lead Backend Engineer (Responsible)<br>- QA Lead (verification) |
| **CM-4 Blockchain contingency (fork/mismatch)** | - Blockchain Architect<br>- DevOps Lead (Accountable/Responsible)<br>- Security/Audit support (consulted) |
| **CM-5 Data recovery & continuity** | - DevOps/Infrastructure Lead (Accountable)<br>- DBA/Backend support (Responsible) |
| **CM-6 Quality & traceability safeguards** | - QA Lead (Accountable)<br>- Requirements Engineer (iRTM updates) |
| **CM-7 Supply chain continuity protection** | - PM + DevOps Lead (Accountable)<br>- Customer Ops rep (consulted for blackout windows) |
| **CM-8 Audit trail preservation** | - Security/Audit Lead<br>- Requirements Engineer (Accountable)<br>- DevOps (log retention controls) |

### 6.3. Architecture Requirements

TR-04 Three-tier architecture
TR-05 Load balancing & auto-scaling
TR-06 Dedicated blockchain validator nodes
TR-07 ASP.NET Core MVC
TR-21 Smart contract language (C#)
TR-22 MSSQL with T-SQL

**6.4. DevOps & Deployment**
TR-29 CI/CD pipeline
TR-30 Blue-green deployment
TR-31 Patch management within 48 hours

**6.5. Migration & Transition**
TR-32 Migration tools
TR-34 Migration validation
TR-35 Rollback plan

**7. SIGN OFF**
This Software Requirements Specification (SRS) has undergone regulated-revision, specification, and validation in accordance with the Requirements Engineering process.
The requirements contained within this document represent the agreed and controlled baseline for the Medicare Plus system.

Upon approval, this SRS shall serve as the authoritative reference for system design, implementation, testing, and acceptance activities. Any modifications to the requirements shall be managed through the formal change control protocol and reflected in the Requirements Traceability Matrix (RTM).

Approval Status: Validated and approved for transition to System Design and Development Phase.

**Customer Representatives**

| Role | Name | Signature | Date |
|---|---|---|---|
| Chief Information Officer | | | |
| Head of Compliance | | | |
| Supply Chain Director | | | |

**Vendor Representatives**

| Role | Name | Signature | Date |
|---|---|---|---|
| Project Manager | | | |
| Lead Requirements Analyst | | | |

**-- END OF REPORT --**

## FIGURES

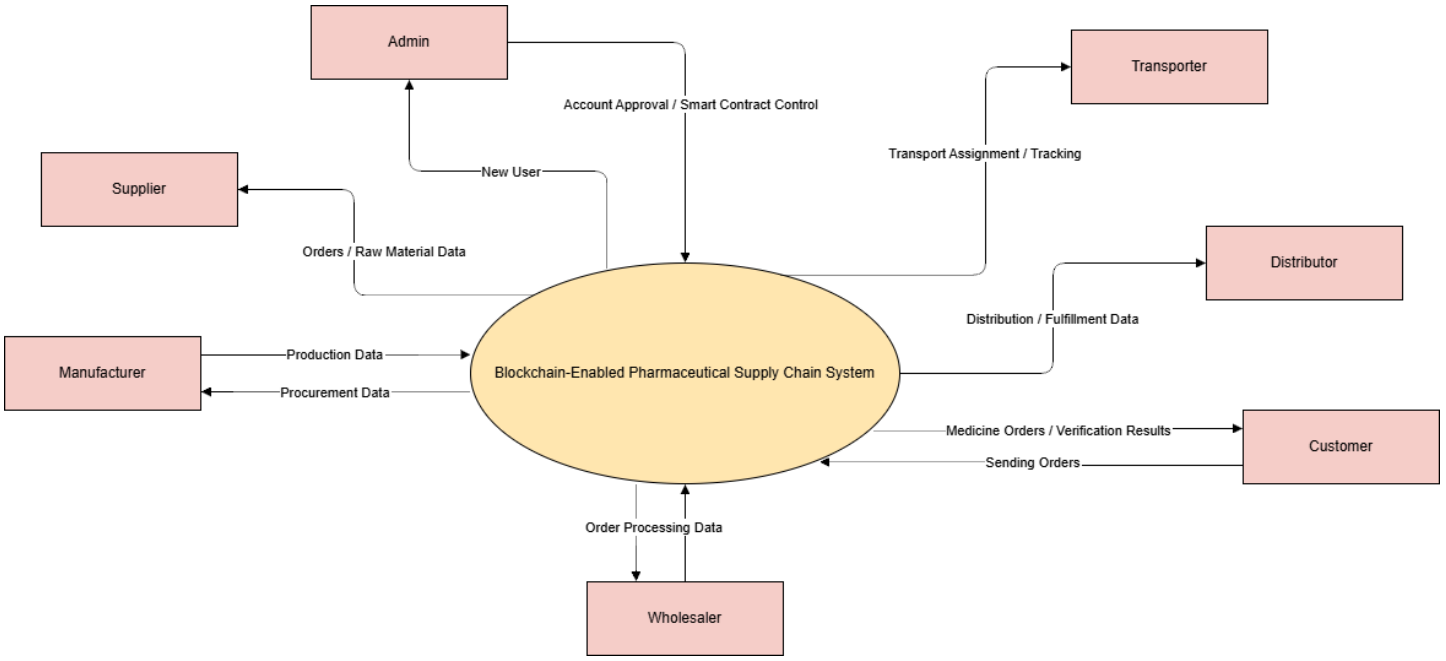


**Fig 1. Context diagram showing data flows of the System.**

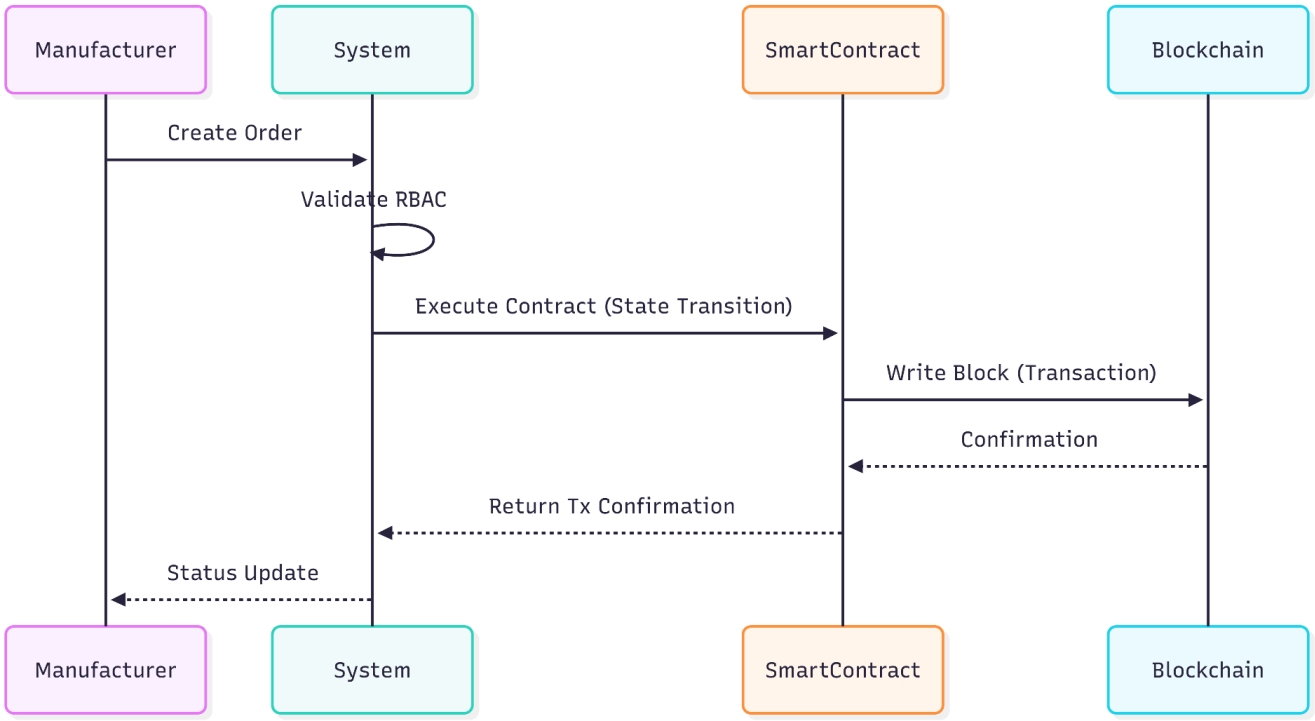


**Fig 2. Sequence Diagram for how the system operates**

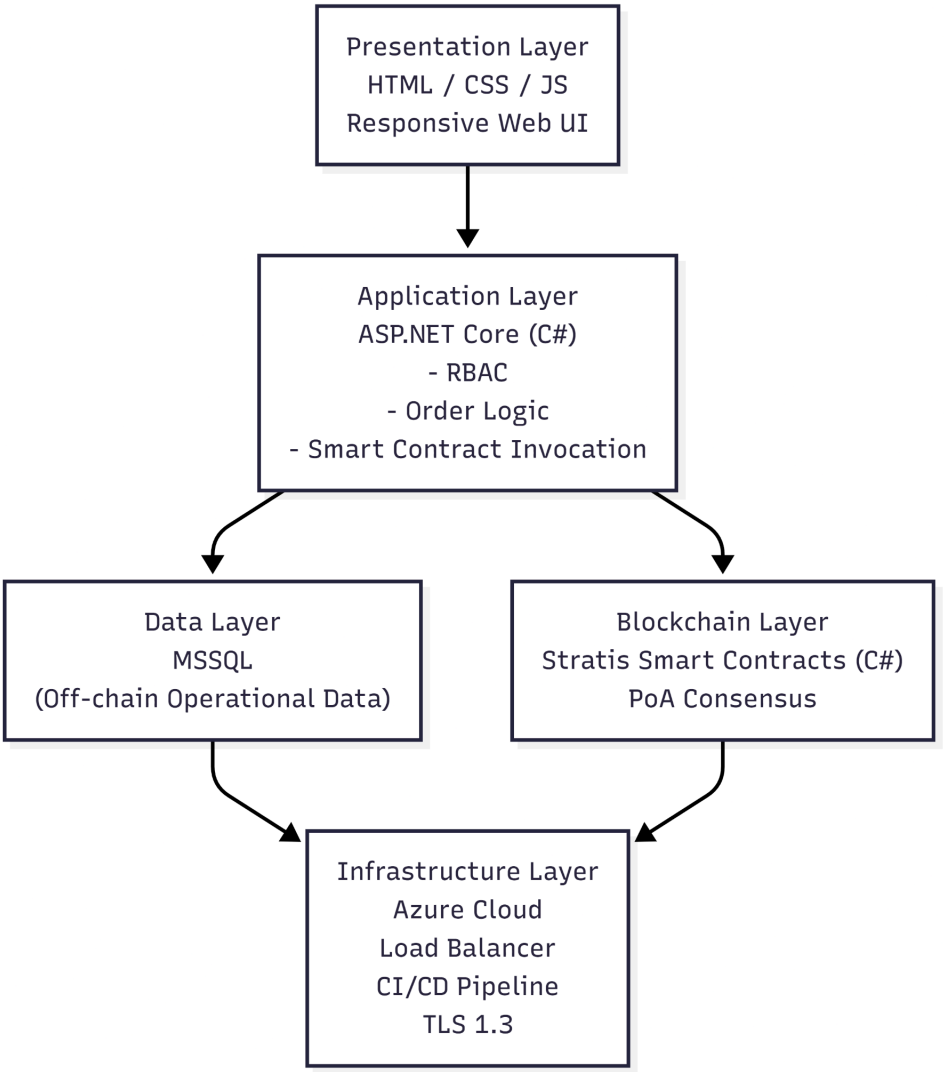


**Fig 3. Architecture Diagram of the system and its components**

## Inceptive Requirements Traceability Matrix (iRTM)

| | |
|---|---|
| **Document type** | Inceptive Requirements Traceability Matrix (iRTM) |
| **Project ID** | BC-0001 |
| **Project name** | Smart Contract Enabled Pharmaceutical Supply Chain System |
| **Document date** | 7th March 2026 |
| **Assumption(s)** | *A-01 to A-04: Technical capabilities (internet, digital literacy, HSA licenses, GPS tracking)*<br>*A-05 to A-06: Infrastructure viability (Azure Singapore, Stratis platform)*<br>*A-07 to A-09: Device support (QR scanning, API stability, product serialization)*<br>*A-10: Regulatory stability during project lifecycle* |
| **Constraint(s)** | *C-01 to C-03: Technology mandates (Azure Singapore, ASP.NET Core 6.0, Visual Studio 2019+)*<br>*C-04 to C-05: Project limitations (fixed budget/timeline, 99.9% SLA, 4hr maintenance windows)*<br>*C-06 to C-08: Performance requirements (30s blockchain confirmation, 95% queries <500ms, 500 concurrent users, max 10 servers)*<br>*C-09 to C-11: Compliance mandates (7-year retention, MOH security guidelines, WCAG 2.1 AA)*<br>*C-12: Phase 2 deferral (temperature monitoring/cold chain)* |
| **Regulation(s)** | *R-01 to R-03: HSA regulations (GDP, GMP, Health Products Act 2007)*<br>*R-04 to R-05: MOH regulations (Healthcare Services Act 2020, Cyber Guidelines 2023)*<br>*R-06: PDPC PDPA 2012 (data protection, 7-year retention, breach notification)*<br>*R-07: IMDA Electronic Transactions Act (smart contract legal validity)*<br>*R-08 to R-09: ISO standards (9001:2015 Quality, 27001:2013 Security)*<br>*R-10: Medical Device Act (if applicable)* |

| Ser | PS | | | | | SRS | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Ser | Category | ID | Description | Stakeholder | Precedence | Category | ID | Description | Owner | Ranking |
| **Blockchain Infrastructure Requirements** | | | | | | | | | | |
| 1 | Biz Ops | PS-BR-01 | Blockchain recording | System Admin | Essential | IT tech | AR-FR-1.1 / FR-1.1 | Record all transactions on blockchain ledger | Dev Team | High |
| 2 | IT tech | PS-BR-02 | Tamper detection | Auditor | Essential | IT tech | AR-FR-1.2 / FR-1.2 | Detect and flag any tamper attempts on chain | Security Team | High |
| 3 | IT tech | PS-BR-03 | Smart contract automation | System Admin | Desirable | IT tech | AR-FR-1.3 / FR-1.3 | Automate business rules via smart contracts | Dev Team | Medium |
| **User & Role Management Requirements** | | | | | | | | | | |
| 4 | Biz Ops | PS-BR-04 | Account creation | Customer / Admin | Essential | Biz Ops | AR-FR-2.1 / FR-2.1 | Create and manage user accounts on platform | Dev Team | High |
| 5 | Biz Ops | PS-BR-05 | Role access control | System Admin | Essential | IT tech | AR-FR-2.4 / FR-2.4 | Assign and enforce role-based access privileges | Security Team | High |
| 6 | Biz Ops | PS-BR-06 | Customer approval | Operations Manager | Essential | Biz Ops | AR-FR-2.3 / FR-2.3 | Updates account status | Operations | High |
| **Order Lifecycle Requirements** | | | | | | | | | | |
| 7 | Biz Ops | PS-BR-07 | Raw material orders | Procurement Officer | Essential | Biz Ops | AR-FR-3.1 / FR-3.1 | Place and track raw material purchase orders | Procurement | High |
| 8 | Biz Ops | PS-BR-08 | Medicine orders | Pharmacist | Essential | Biz Ops | AR-FR-3.2 / FR-3.2 | Place and track medicine purchase orders | Pharmacist | High |
| 9 | IT tech | PS-BR-09 | Status update | Logistics Officer | Desirable | IT tech | AR-FR-3.4 / FR-3.4 | Update and broadcast order status in real-time | Logistics | Medium |
| 10 | Biz Ops | PS-BR-10 | Transport assignment | Logistics Officer | Essential | Biz Ops | AR-FR-3.5 / FR-3.5 | Assign transport provider to each order leg | Logistics | High |
| **Product Verification Requirements** | | | | | | | | | | |
| 11 | IT tech | PS-BR-11 | Source lookup | Quality Inspector | Essential | IT tech | AR-FR-4.1 / FR-4.1 | Look up origin and provenance of any product | QA Team | High |
| 12 | Biz Ops | PS-BR-12 | Audit display | Auditor | Desirable | IT tech | AR-FR-4.2 / FR-4.2 | Display full audit trail for product on-screen | Auditor | Medium |
| **Non-Functional Requirements (Security & Compliance)** | | | | | | | | | | |
| 13 | IT tech | PS-TR-01 | Encryption | Security Officer | Essential | IT tech | AR-(NFR-9& NFR-10) | All data at rest and in transit must be encrypted | Security Team | High |
| 14 | IT tech | PS-TR-02 | Hash algorithm | Security Officer | Essential | IT tech | AR-NFR-14 / NFR-14 | Use approved cryptographic hash for block linking | Security Team | High |
| 15 | IT tech | PS-TR-03 | Performance | System Admin | Desirable | IT tech | AR-(NFR-1 - NFR-7) | System shall process transactions within SLA time | Dev Team | Medium |
| 16 | IT tech | PS-TR-04 | Uptime | Operations Manager | Essential | IT tech | AR-NFR-15 / NFR-15 | System shall meet required availability uptime % | Ops Team | High |
| 17 | Biz Ops | PS-TR-05 | Compliance logs | Compliance Officer | Essential | Biz Ops | AR-NFR-12 / NFR-12 | Maintain tamper-proof compliance audit log records | Compliance | High |

**Fig 4. Inceptive Requirements Traceability Matrix**

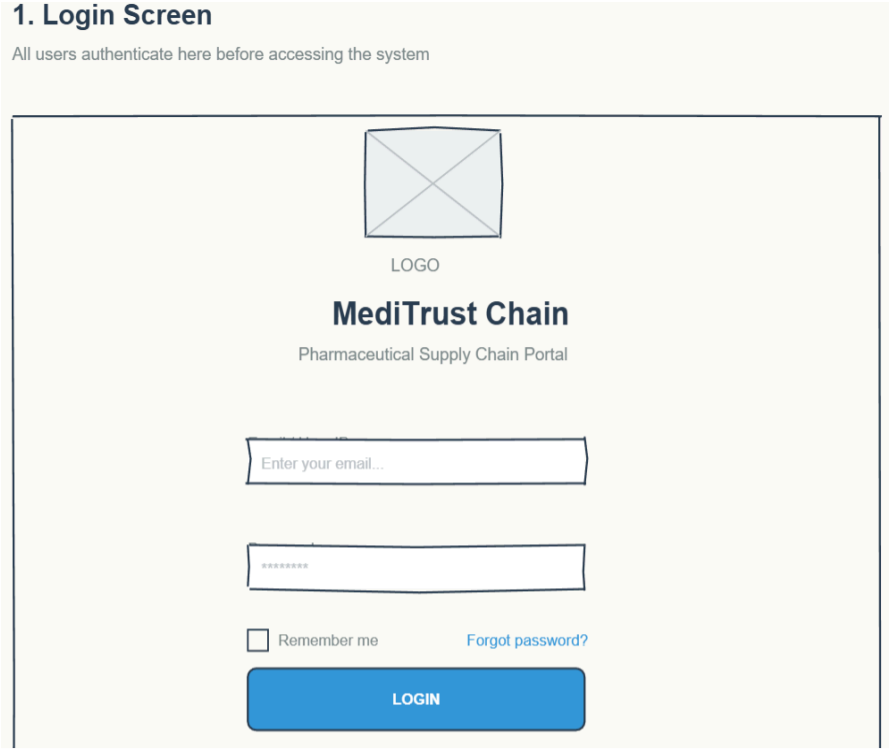


**Fig. 5.1 Wireframe Login Screen**

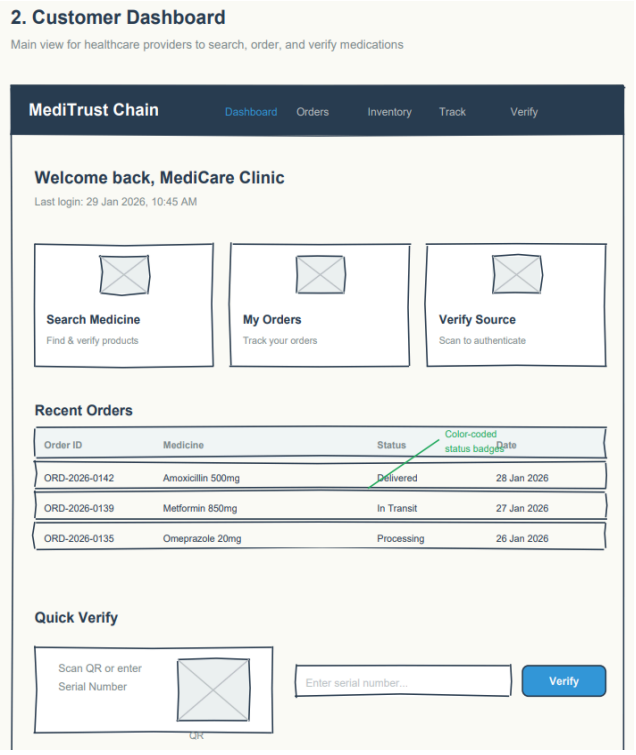


**Fig. 5.2 Wireframe Customer Dashboard**

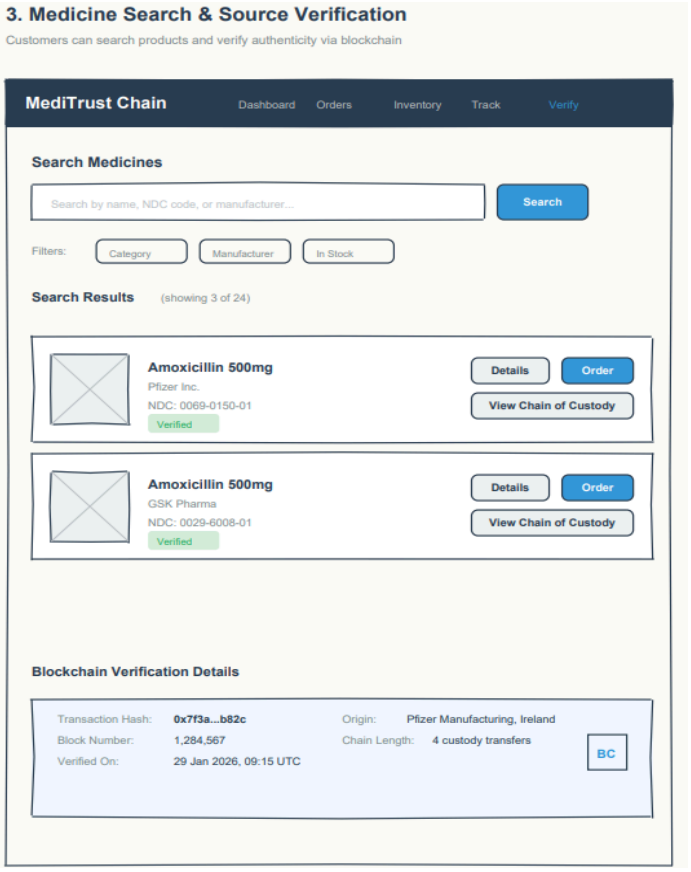


**Fig. 5.3 Wireframe Medicine Search & Source Verification**

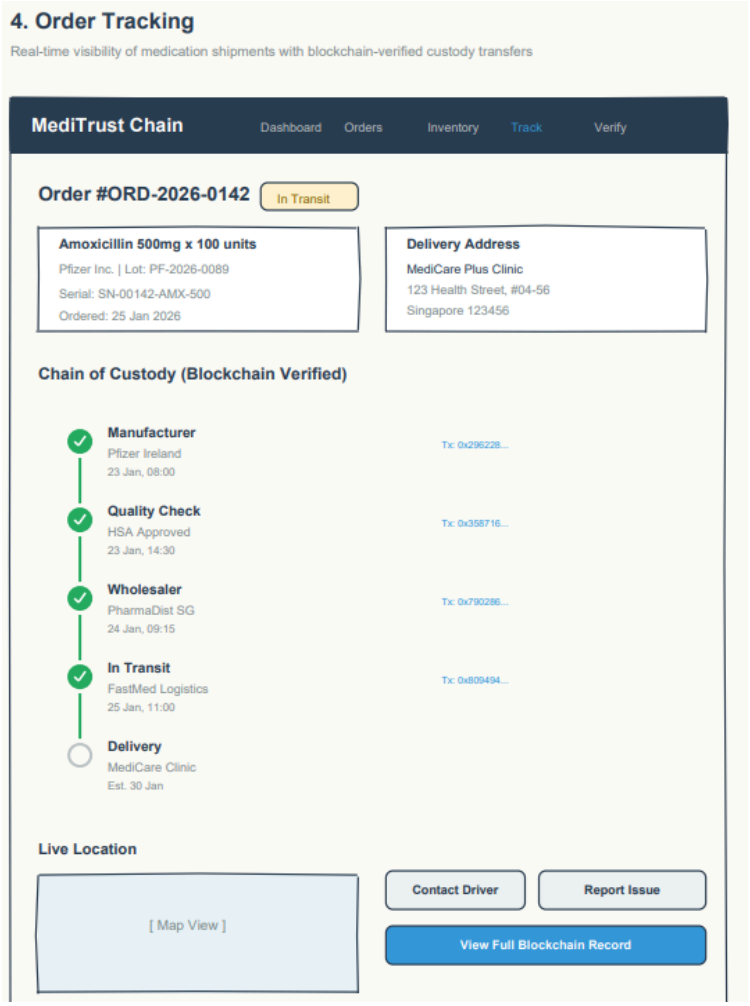


**Fig. 5.4 Wireframe Order Tracking**

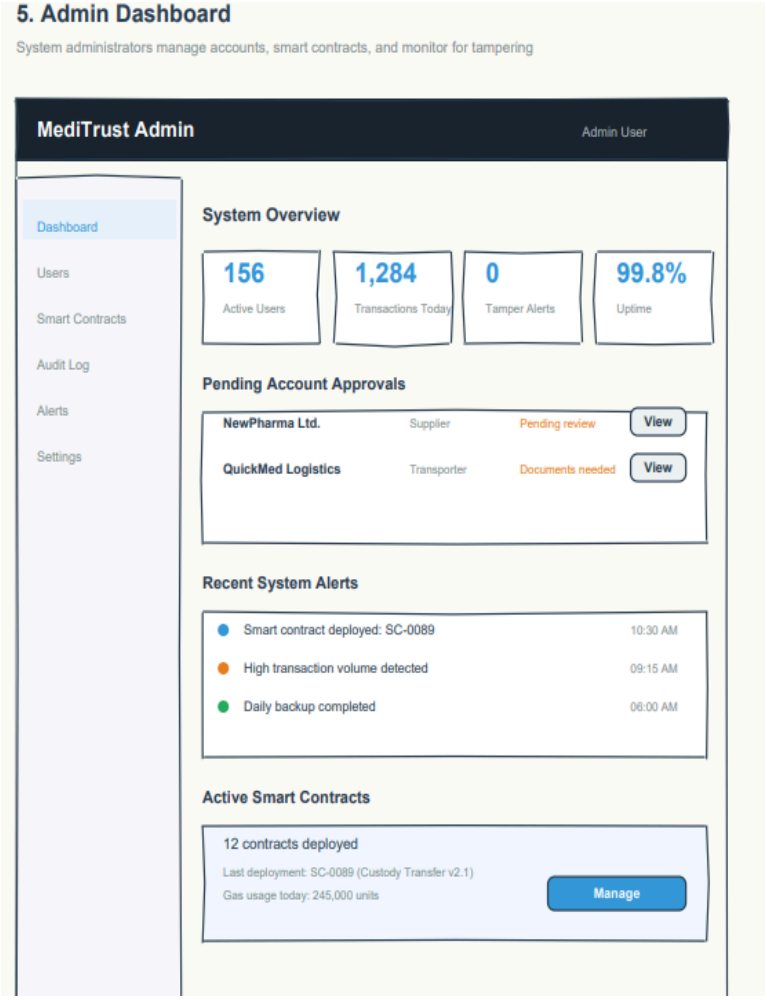


**Fig. 5.5 Wireframe Admin Dashboard**

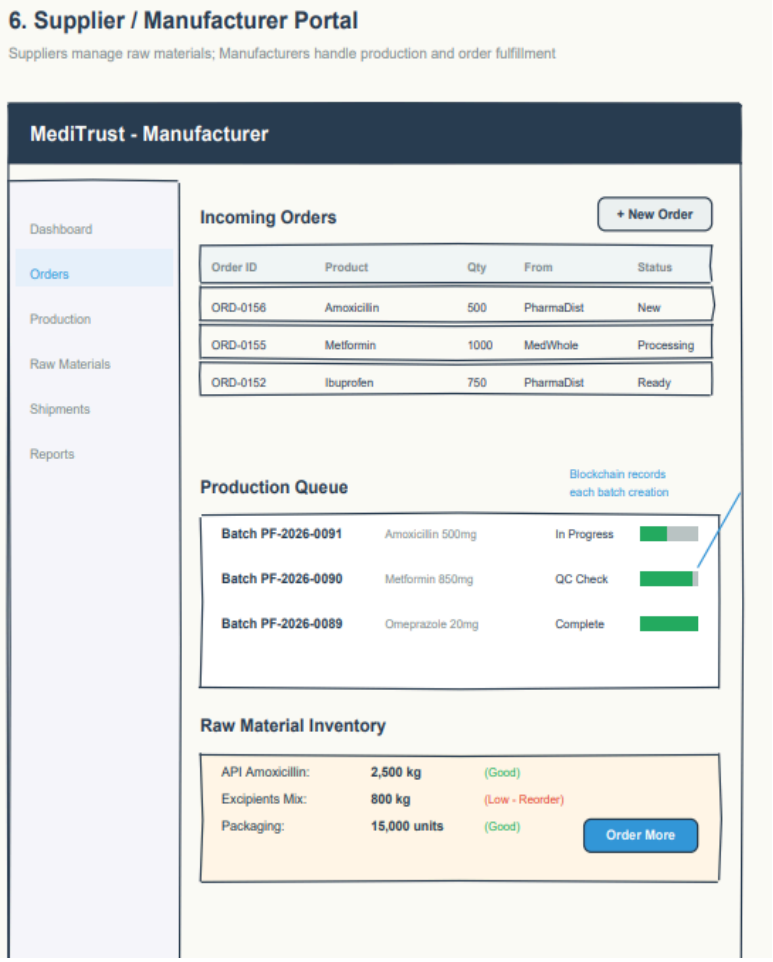


**Fig. 5.6 Wireframe Supplier/Manufacturer Portal**